

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING

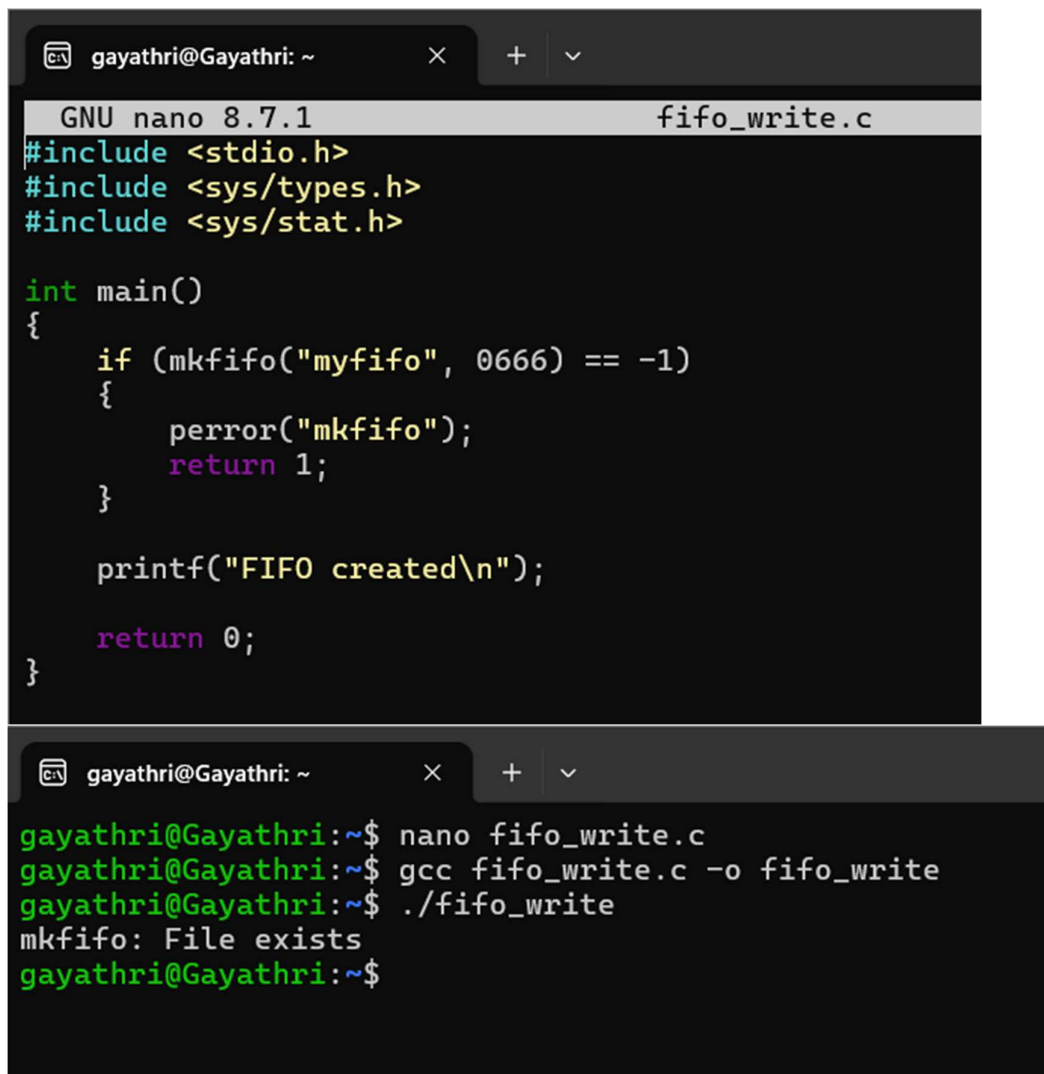
Sec:- 2

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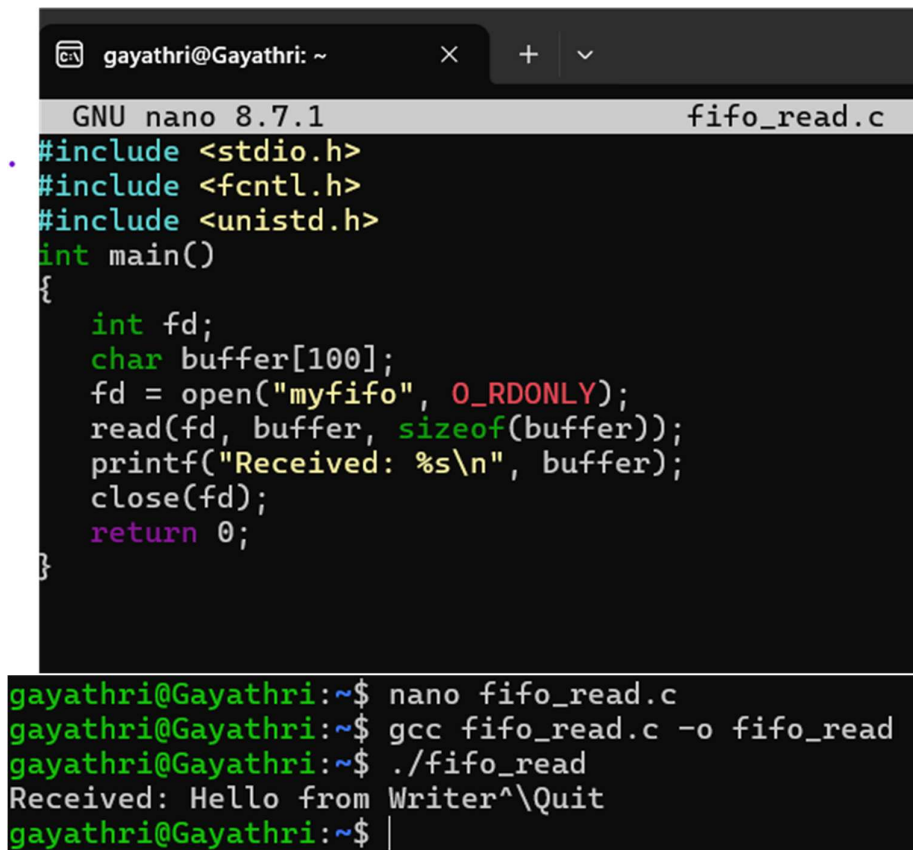
Pipes

1. Create FIFO



```
gayathri@Gayathri: ~  
GNU nano 8.7.1 fifo_write.c  
#include <stdio.h>  
#include <sys/types.h>  
#include <sys/stat.h>  
  
int main()  
{  
    if (mkfifo("myfifo", 0666) == -1)  
    {  
        perror("mkfifo");  
        return 1;  
    }  
  
    printf("FIFO created\n");  
  
    return 0;  
}  
  
gayathri@Gayathri:~$ nano fifo_write.c  
gayathri@Gayathri:~$ gcc fifo_write.c -o fifo_write  
gayathri@Gayathri:~$ ./fifo_write  
mkfifo: File exists  
gayathri@Gayathri:~$
```

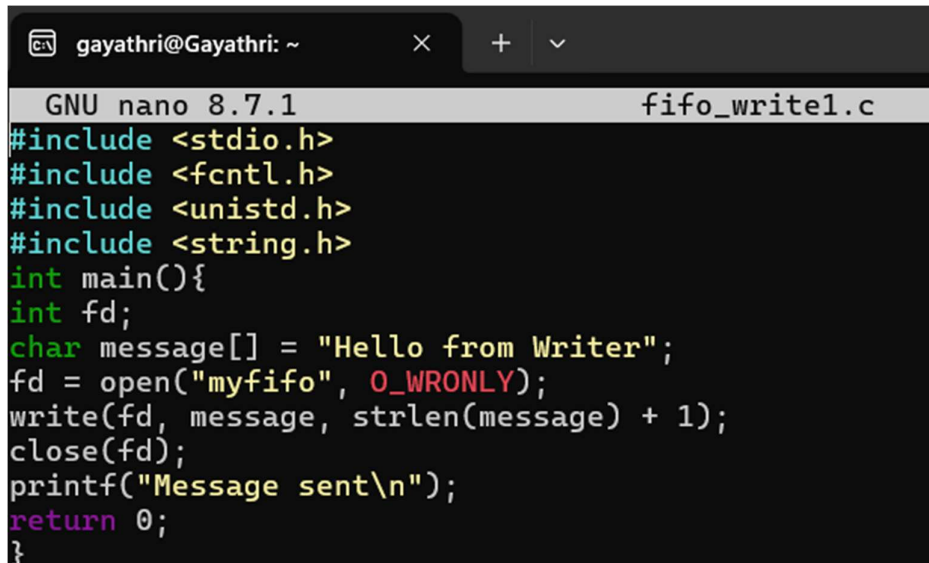
2. Read from FIFO



The image shows a terminal window with a dark background. The top part of the window displays the GNU nano 8.7.1 editor editing a file named `fifo_read.c`. The code inside the editor is a C program that opens a FIFO named `myfifo` in read-only mode, reads data into a buffer, and prints it. The bottom part of the window shows the command prompt where the program is compiled with `gcc` and executed. The output shows the program receiving data from a writer and printing it.

```
gayathri@Gayathri: ~  
GNU nano 8.7.1 fifo_read.c  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
int main()  
{  
    int fd;  
    char buffer[100];  
    fd = open("myfifo", O_RDONLY);  
    read(fd, buffer, sizeof(buffer));  
    printf("Received: %s\n", buffer);  
    close(fd);  
    return 0;  
}  
  
gayathri@Gayathri:~$ nano fifo_read.c  
gayathri@Gayathri:~$ gcc fifo_read.c -o fifo_read  
gayathri@Gayathri:~$ ./fifo_read  
Received: Hello from Writer^\Quit  
gayathri@Gayathri:~$ |
```

3. Write to FIFO



```
gayathri@Gayathri: ~  
GNU nano 8.7.1 fifo_writel.c  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
#include <string.h>  
int main(){  
int fd;  
char message[] = "Hello from Writer";  
fd = open("myfifo", O_WRONLY);  
write(fd, message, strlen(message) + 1);  
close(fd);  
printf("Message sent\n");  
return 0;  
}
```